IV B.TECH – I SEMESTER

Building Materials And Services

Open Elective-III | Code: 23A01705a | JNTUA R23 · B.Tech CSE

Course Code	BUILDING MATERIALS AND SERVICES (OPEN ELECTIVE – III)	L	T	P	C
23A01705a		3	0	0	3
Course Objectives: **The objectives of this course are to make the student :** To understand the properties, classifications, and applications of building materials like stones, bricks, tiles, wood, aluminum, glass, paints, and plastics. To analyze the composition, manufacturing process, and properties of cement and admixtures. To apply knowledge of building components such as lintels, arches, walls, stairs, floors, roofs, foundations, and joinery. To evaluate masonry, mortars, finishing techniques, and formwork systems. To assess various building services including plumbing, ventilation, air conditioning, acoustics, and fire protection.					
Course Outcomes: **Upon successful completion of the course, students will be able to:** Understand the properties, classifications, and applications of building materials like stones, bricks, tiles, wood, aluminum, glass, paints, and plastics. Analyze the composition, manufacturing process, and properties of cement and admixtures. Apply knowledge of building components such as lintels, arches, walls, stairs, floors, roofs, foundations, and joinery. Evaluate masonry, mortars, finishing techniques,					

and formwork systems. Assess various building services including plumbing, ventilation, air conditioning, acoustics, and fire protection.					
CO – PO Articulation Matrix **Course Outcomes PO1 PO2 PO3 PO4 PO5 PO6 PO7 PO8 PO9 PO 10 PO 11 PO 12 PSO1 PSO2 CO -1** 3 - - - 2 - - - - - - - 3 3 **CO -2** 3 3 - - 2 - - - - - - 2 3 3 **CO -3** 3 - 3 2 3 - - - - - - - - 3 3 **CO -4** - - 3 3 3 - 2 - - - - - 3 3 **CO -5** - - - - - 3 3 2 - - - - - 3					
UNIT – I					
StonesandBricks, Tiles: Building Stones – Classifications and Quarrying – Properties – Structural Requirements – Dressing. Bricks – CompositionofBrick Earth – Manufacture and Structural Requirements, Fly Ash, Ceramics. Timber, Aluminum, Glass, PaintsandPlastics: Wood - Structure – Types and Properties – Seasoning – Defects; Alternate Materials for Timber – GI / Fibre – Reinforced Glass Bricks, Steel & Aluminum, Plastics.					
UNIT – II					
Cement &Admixtures: types of Cement - Ingredients of Cement – Manufacture – Chemical Composition – Hydration - Field & Lab Tests – Fineness – Consistency – Initial &Final Setting – Soundness . Admixtures – Mineral & Chemical Admixtures – Uses					
UNIT – III					
Building Components: Lintels, Arches, Walls, Vaults – Stair Cases – Types of Floors, Types of Roofs – Flat, Curved, Trussed; Foundations – Types; Damp Proof Course; Joinery – Doors – Windows – Materials – Types.					
UNIT – IV					

UNIT – IV					
Mortars, MasonryandFinishing's Mortars: Lime and Cement Mortars Brick Masonry – Types – Bonds; Stone Masonry – Types; Composite Masonry – Brick-Stone Composite; Concrete, Reinforced Brick. Finishers: Plastering, Pointing, Painting, Claddings – Types – Tiles – ACP.form Work: Types: Requirements – Standards – Scaffolding – Design; Shoring, Underpinning.					
UNIT – V					
Building Services: Plumbing Services: Water Distribution, Sanitary – Lines &Fittings; Ventilations: Functional Requirements Systems of Ventilations. Air-Conditioning - Essentials andTypes; Acoustics – Characteristic – Absorption – Acoustic Design; Fire Protection – Fire Hazards – Classification of Fire Resistant Materials and Constructions.					
TEXT BOOKS:					
Building Materials and Construction – Arora&Bindra, Dhanpat Roy Publications. Building Materials and Construction by G C Sahu, Joygopal Jena McGraw hill Pvt Ltd 2015.					
REFRENCE BOOKS:					
Building Construction by B. C. Punmia, Ashok Kumar Jain andArun Kumar Jain - Laxmi Publications (P) ltd., New Delh P. C. Varghese, Building Materials, Prentice Hall of India, 2015. N.Subramanian ,"Building Materials Testing and Sustainability", Oxford Higher Education, 2019. R. Chudley, Construction Technology, Longman Publishing Group, 1973. S. K. Duggal, Building Materials, Oxford & IBH					

Publishing Co. Ltd., New Delhi, 2019					
Online Learning Resources:					
https://archive.nptel.ac.in/courses/105/102/105102088/					